

QA Spectrophotometer

2025-LGD-0129

1. OVERVIEW / BACKGROUND. The Library of Congress (Library) requires contract support to purchase a spectrophotometer for use in its quality assurance program for the testing of papers, boards, etc. to multiple ISO standards. Testing the physical and chemical properties of materials is central to the Library's quality assurance testing program for materials used in the storage and housing of collections as well as multiple ongoing research programs. Many of the specifications rely on published ISO testing standards such that specific capabilities and geometries of testing equipment are required as they are specified in the international standards (ISOs) that have been used to develop the Library's specifications.
2. SCOPE. The Preservation Research and Testing Division (PRTD) requires a spectrophotometer for use in its quality assurance program for the testing of papers, boards, etc. to multiple ISO standards. This instrument will replace extant equipment that is no longer supported and has exceeded its useful lifespan. Testing the physical and chemical properties of materials is central to the Library's quality assurance testing program for materials used in the storage and housing of collections as well as multiple ongoing research programs. The contract shall include all parts and equipment to meet specified parameters, installation, user training, shipping, and 1 year of warranty support.
3. REQUIREMENTS. A benchtop spectrophotometer system that includes:
 - 3.1 A benchtop spectrophotometer system that meets or exceeds the following specifications:
 - 3.1.1 Compliance with ISO Standards 2469, 2470, 2471, 3688, 5631, 11475, 11476, 12625.
 - 3.1.2 Optical geometry utilizing dual beam d/0° arrangement with CIE 2° and CIE 10° detectors.
 - 3.1.3 Pulsed Xenon light source with greater than 500,000 measurement expected lifetime.
 - 3.1.4 Multiple UV sources and filters present to control illuminant specified.
 - 3.1.5 Multiple illuminant capabilities including A, B, C, D50, D55, D65, D75, F2, F7, F11.
 - 3.1.6 Standard aperture with specimen measurement area of 30mm supplied with additional apertures available for purchase with smaller specimen measurement areas.
 - 3.1.7 Automated calibration capability.
 - 3.1.8 User selectable reflectance spectral intervals of better than 5nm across full spectrum of 400nm-700nm.
 - 3.1.9 Repeatability greater than <0.01 CIELAB DE with white tile reference.
 - 3.1.10 Capable of measuring from samples from 1um to 40mm in thickness.

3.1.11 Measurement results produced from reflectance measurements for Brightness, Whiteness (CIE, ASTM, Hunter, etc.), Yellowness, Tint, Color (XYZ/Rx, Ry, Rz/L,a,b/L*,a*,b*/DWL, PUR), Color Difference, Fluorescence, Opacity, full spectral data from 400-700 nm.

3.2 Installation and a day of training/system introduction are included.

3.2.1 All costs for engineer travel and housing are included.

3.2.2 Installation timing to be scheduled upon the receipt of the instrument and coordinated with the COR.

3.3 All shipping and handling costs to the Library of Congress are to be included.

4. CONTRACT PERFORMANCE

4.1. SCHEDULE OF EVENTS/DELIVERABLES

Paragraph reference	Description of Event/Deliverable	Due date/Payment schedule
3.1	Delivery of spectrophotometer and all standard accessories.	Within 8 weeks of POP.
3.2	On-site installation of equipment and user training.	Within 3 weeks of delivery.
4.2	One-year warranty on the product that includes all parts, labor, and shipping	After official acceptance and its installation. Period of Performance one year.

4.2. WARRANTY (If applicable). The contractor shall provide a one-year warranty on the product that includes all parts, labor, and shipping valid for one year from the date of installation of the part or repair.

5. ADDITIONAL INFORMATION

5.1. PLACE OF PERFORMANCE. The work to be performed under the contract will be performed at Library of Congress, Preservation Research and Testing Division, 101 Independence Ave. SE, Madison Building Room LM-G52, Washington DC, 20540.

5.2. HOURS OF OPERATION. The contractor is responsible for conducting business between the hours of 8:30 am to 5:00 pm Mondays through Fridays, except Federal holidays or when the Government facility is closed due to local or national emergencies, administrative closings, or similar Government directed facility closings.

5.3. Government furnished property. Vendor may use consumables available in the laboratory that are compatible with the obtained instrumentation.

(End of Performance Work Statement / Statement of Work)